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On-the-fly parameter estimation based on item response theory in item-based adaptive learning systems

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Abstract
Online learning systems are able to offer customized content catered to individual learner's needs, and have seen growing interest from industry and academia alike in recent years. In contrast to the traditional computerized adaptive testing setting, which has a well-calibrated item bank with new items added periodically, the online learning system has two unique features: (1) the number of items is large, and they have likely not gone through costly field testing for item calibration; and (2) the individual's ability may change as a result of learning. The Elo rating system has been recognized as an effective method for fast updating of item and person parameters in online learning systems to enable personalized learning. However, the updating parameter in Elo has to be tuned post hoc, and Elo is only suitable for the Rasch model. In this paper, we propose the use of a moment-matching Bayesian update algorithm to estimate item and person parameters on the fly. With sequentially updated item and person parameters, a modified maximum posterior weighted information criterion (MPWI) is proposed to adaptively assign items to individuals. The Bayesian updated algorithm along with MPWI is validated in a simulated multiple-session online learning setting, and the results show that the new combo can achieve fast and reasonably accurate parameter estimations that are comparable to random selection, match-difficulty selection, and traditional online calibration. Moreover, the combo can still function reasonably well with as low as 20% of items being pre-calibrated in the item bank.

Keywords Item calibration · Item bank construction · Adaptive learning system · Elo rating system · Bayesian

Introduction

Modern technologies have made computer-assisted learning increasingly popular. Incorporating a computerized component into a learning system allows for innovative procedures to facilitate the learning process. One of the major benefits of these online learning systems is adaptivity. Compared with traditional classroom instruction where every learner receives the same materials, performs the same tasks, and solves the same problems, a computerized adaptive learning system is able to provide customized course content, format, and practice sessions to match the needs of individual learners (Wauters et al., 2010). The same idea can also be applied to the practice and problem-solving component of the online learning systems. Recently, there has been an increasing amount of research (e.g., Conejo et al., 2016; Chrysafiadi et al., 2020) dedicated to the development of such systems that deliver personalized items (sometimes with hint and/or feedback) to learners in the hope that students would gain knowledge and improve their ability through interacting with these items. To this end, the system is typically expected to be able to judge the learners' ability level (and sometimes the difficulty of practice items), and adaptively present items that can optimize the individual learning process.

Although the development of these online practice systems is still in an early phase, there has been some research effort to transfer the expertise from educational testing to the learning environment. Computerized adaptive testing (CAT) is a well-established procedure widely used in the testing industry and is very similar to adaptive learning systems in many ways. At the core, both CAT and adaptive practice systems deliver items adaptively according to students' ability

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levels and involve similar procedures such as item selection and ability estimation. Despite the similarities, these two systems still differ in some important respects due to their distinct nature of testing and learning. For instance, in a learning environment it is expected that students' ability would show some form of change (preferably improvement) over time, while in testing it is assumed that ability will remain constant throughout the testing session. Another difference is that a learning program requires many more items for a single user and hence needs a larger item bank. These differences pose technical and practical challenges when applying CAT methodology in the learning environment (Eggen, 2012; Veldkamp et al., 2011; Wauters et al., 2010). This paper intends to address the item bank construction challenge by adapting a recently proposed Bayesian update rule and proposing a new item assignment strategy. The Bayesian update algorithm is easy to implement with closed-form equations that can quickly provide real time parameter updates. Along with a new item assignment strategy, this combo is evaluated in an online learning setting where abilities change across learning sessions, and in some extreme scenarios where the item bank only has 20% of items being pre-calibrated. Note that due to the changing abilities in a learning context, and due to the inherent scale indeterminacy of item response theory (IRT), some pre-calibrated items are necessary to anchor the scale. We consider 20% known anchor items as a reasonable and much more affordable assumption than what is normally considered in an online calibration setting, where 80% or even 90% are often considered as known operational items (e.g., Chen et al., 2012, 2017; Chen & Wang, 2016).

Item bank construction

In CAT, an item bank consists of items with known parameters that are obtained from previous calibration studies. The process of item calibration in IRT is known to be costly and time-consuming, as every item in the bank needs to be administered to a representative and motivated sample of adequate size (Wauters et al., 2010). This requires large, high-quality datasets that are oftentimes only affordable for large-scale, high-stakes testing programs (Conejo et al., 2016). Although the procedure for the item parameters used in learning applications may not need to be as rigorous as in testing (Veldkamp et al., 2011), parameter estimates are still needed in order to achieve adaptivity. Item calibration in the learning environment can be particularly challenging because more items are included in practice and instruction than in testing due to the former having fewer constraints on time and test length.

Researchers have adopted different approaches to address the issue of item bank calibration. One way is to bypass the calibration phase and derive item parameter estimates from other sources. Some studies made use of human judgment, such as feedback from domain experts, teachers, or learners themselves, to provide estimates for item difficulty. (e.g., Conejo et al., 2014; Wauters et al., 2012). Their results showed that these estimates tended to be less reliable than calibration based on IRT models and could be equally resource-demanding in the face of a large item bank. Others proposed to construct statistical models that can predict the difficulty of items based on item features (Matteucci et al., 2012). For instance, in a mobile app that offers training in English vocabulary, the difficulty of the individual words in the system was estimated from a weighted combination of word properties such as word length, number of phonemes, and their frequency of usage (Chen & Chung, 2008). Note that building these predictive models usually requires justification from substantive knowledge, which may not exist in some fields. Additionally, parameter estimates obtained from this statistical analysis are only preliminary and must be updated after operational data start to accumulate.

Another line of research borrows from the idea of online calibration, which is a useful procedure in CAT that calibrates new items over the course of a testing program. Specifically, CAT system would select several new items and deliver these items randomly in operational tests. The responses to new items are usually not used for scoring, since the parameters of new items are either unknown or unstable during the testing process; instead they are collected to calibrate or update the parameters of new items (van der Linden & Ren, 2015; Zheng, 2016; Chen et al., 2012; Chen et al., 2017).

Online calibration is useful in calibrating new items when they are being developed and added over time, but it doesn't really alleviate the difficulty of item bank development at the initial phase. There has been recent research that modified the procedures of online calibration into a strategy called "continuous calibration," which allowed CAT to proceed without a full-fledged operational bank (Born et al., 2019; Fink et al., 2018; Makransky & Glas, 2010). During the beginning stages the test starts with minimal, if any, adaptivity, and essentially serves as a makeshift calibration study. This provides some preliminary parameter estimates. Then, in later stages when there are more and better calibrated items in the bank, the test becomes increasingly adaptive as more items with trustworthy parameter estimates become amenable to item selection. One of the issues of applying the strategy to the learning environment is that they are too slow in moving from random stages to adaptive stages. There are clear-cut boundaries between these stages, and item parameter estimates are not available until they enter the adaptive stages. Therefore, in the first few random stages, even if items have been delivered many times, they are still not selected adaptively. Also, because the new items require a relatively large calibration sample to be ready, there are

usually not enough operational items to choose from in the first few adaptive stages, and so they are not as adaptive as intended. The system only functions as a fully adaptive process at a very late stage into the test, and this would likely lead to suboptimal measurement performance.

Elo rating system (ERS)

Many of the existing studies on item-based adaptive learning systems made use of the Elo rating system (ERS; Elo, 1978) as the underlying model (Pelánek, 2016). ERS can be deemed a special case of the paired-comparison models in the statistics literature (for a review, see Cattelan, 2012) and was originally proposed to measure the underlying ratings of chess players using results from chess games. The system is dynamic in the sense that players' ratings can be updated after every match (or set of matches, such as a tournament), making it an ideal choice for adaptive online learning where tracking the changes in learner ability on the fly is of primary concern.

The basic idea of the ERS is to calculate the expected probability of game outcome based on the current ratings of the two participating sides and update the ratings according to the difference between the expected probability and actual observed outcome. The winning player receives a higher rating after the update, and the update size depends on the rating difference between the two players. Consider the simplest scenario where there are no ties and a game between player i and j has only two possible outcomes: $u_{ij}=1$ when player i wins versus player j and $u_{ij}=0$ when player i loses versus player j. Let θ_i and θ_j be the current ratings of the two players and p_{ij} the expected probability of player i defeating player j; then both players' new ratings after the game are given by

$$\theta_i' = \theta_i + K\left(u_{ij} - p_{ij}\right) \tag{1}$$

$$\theta_j' = \theta_j - K\left(u_{ij} - p_{ij}\right) \tag{2}$$

The value K is a weighting factor that can either be a constant or estimated from data. To adapt the system to educational applications, one can consider a chess game between two players as analogous to an interaction between an item and a learner. Then one can rewrite θ_j in Eq. 2 as the difficulty parameter b_j of item j. This also makes it possible in theory to estimate ability and item parameters at the same time.

Current research on item-based adaptive learning systems

As mentioned in the previous section, much of the existing research on adaptive learning systems employed different variations of ERS to track learner ability on the fly. One of the first comprehensive studies on item-based adaptive learning systems was conducted by Klinkenberg et al. (2011), in which they provided technical details on what they called the computerized adaptive practice (CAP) system and proposed a modified version of ERS. Several follow-up studies explored different aspects of the CAP system. Wauters et al. (2012) investigated item parameter calibration and compared the item difficulty estimates derived from ERS with those from the Rasch model. It was concluded that ERS and the IRT estimation did not differ greatly in terms of calibration accuracy with sample sizes varying from 20 to 300. Note that, similar to IRT, ERS also needs to fix the scale arbitrarily (i.e., setting the mean to 0) and/or include pre-calibrated items during the calibration. Another extension of the ERS-based learning system adopted the explanatory IRT modeling approach to provide better estimates of the starting values of learner ability by incorporating their previous learning history and background information (Park et al., 2019a). More recently, a multidimensional extension of ERS in CAP environments was proposed to monitor multiple abilities simultaneously (Park et al., 2019a). The results showed a further improvement in ability estimation accuracy.

The ERS-based approach provides practically feasible solutions to many of the challenges encountered in building an adaptive learning system. According to Pelánek (2016), the system "*is particularly attractive when we want to build a reasonably behaving system quickly and cheaply*" (p. 177). It is also fast to compute, which makes it suitable for the tasks of real-time parameter estimation and dynamic ability tracking. All these benefits, however, come at the cost of compromised measurement accuracy, due to the simplistic nature of the model.

ERS is mathematically similar to the Rasch model, which is the simplest IRT model that does not allow items to vary in discriminating power. But the bigger issue is the arbitrariness in setting the update weight K. While the Rasch model uses conditional maximum likelihood to estimate its parameters, ERS does it through an update rule that relies heavily on the choice of K, which directly determines the update size. For ability levels, a large value of K means that the estimates would constantly fluctuate without converging to a certain point, while a small value means it would take too many items to close in on the true value (assuming the ability is constant over time) or to catch up to the ability change that is taking place. Ideally, K should be estimated from data as well, but in all applications thus far, K was either fixed before data collection or computed by some linear equations that included many tuning parameters chosen based on pilot simulation or previous research. As the optimal values of K are likely to vary across different situations, there is no way of knowing whether the choice of K is appropriate until some post hoc analysis is done. As a result, the estimation

methods in such systems are much less based on statistical grounds than those used in IRT research, and the resulting estimates are not as accurate and trustworthy in comparison.

In this paper, an improved update algorithm under the Bayesian framework (Weng & Coad, 2018) is proposed as an alternative. While retaining all the advantages of ERS, the new algorithm includes a variance component for each parameter to quantify the amount of uncertainty around its estimate, and the update size is now data-driven and computed in real time using the current responses and parameter estimates. The Bayesian algorithm can also estimate item discrimination so that now each item could have its own discrimination parameter, which is more flexible than ERS. Furthermore, the new algorithm requires a small number of pre-calibrated items in the adaptive learning system, which greatly reduces the cost of item bank construction. With sequentially updated item and person parameters, a modified maximum posterior weighted information criterion (MPWI) is proposed to adaptively assign items to persons, which takes into account the uncertainty of both sets of parameters.

The structure of the paper is outlined as follows. First, we will introduce the Bayesian update algorithm, followed by the introduction of MPWI as an item assignment criterion. Then the performance of this combo is evaluated by a simulation study in the following section. The final section summarizes the main findings as well as limitations in the simulations and lays out possible directions for future research.

Method

Bayesian update rule

The Bayesian update rules are based on the work by Weng and Coad (2018), which, similar to ERS, stems from the research on the analysis of dynamic paired comparison data in the statistics literature. In the original ERS, the abilities were assumed to be point estimates. Bayesian extension of ERS was later developed to incorporate the precision or variability of the ability estimates as well (Fahrmeir & Tutz, 1994; Glickman, 1993). In this formulation, ability was normally distributed with time-varying means and variance. Instead of being fixed a priori, the update weight K was dynamically calculated from the observed outcome and the previous parameter estimates and their variances. If the ability estimate contains substantial uncertainty (for example, a new student just enters the learning system or there is a long interval since the student's previous activities), the next outcome would have a big impact on the ability update. When there is little certainty about the ability level, the outcomes would not affect the rating too much.

These Bayesian models were estimated offline with either MCMC (Glickman, 1993) or empirical Bayes (Fahrmeir & Tutz, 1994), and they become unwieldy when the number of players or outcomes is large. A noniterative algorithm with closed-form equations for the posterior means and variance that allows for online data processing was derived by approximating the posterior distributions of ability (Glickman, 1999). The algorithm was the basis of the Glicko rating system that has been used extensively in chess and other competitive online games. Further extensions of the Bayesian paired comparison models include the TrueSkill rating system developed by Microsoft for their Xbox™ consoles that could model data from matches of more than two players (Herbrich et al., 2007). Although TrueSkill utilized a deterministic algorithm to approximate the posterior distribution, it is still an iterative procedure and leaves room for further simplification like the closed-form computational formulas.

Weng and Lin (2011) first proposed such a closed-form update formula based on moment matching for paired comparison models. Weng and Coad (2018) later adapted them for IRT models. The general steps of their derivation are summarized below. Suppose ξ_i is a model parameter with prior distribution of $N\left(\mu_i, \sigma_i^2\right)$ and let

$$z_i = \frac{\xi_i - \mu_i}{\sigma_i} \tag{3}$$

The posterior distribution of the parameter vector $z = (z_1, z_2, \ldots z_i, \ldots z_p)^T$ can be written as

$$f(z|\boldsymbol{u}) = \frac{\phi(z) f(\boldsymbol{u}|z)}{\int \phi(z) f(\boldsymbol{u}|z) dz}, \tag{4}$$

where $f(\mathrm{u}|\mathrm{z})$ is the likelihood function of response vector u and ϕ denotes the density of a p-variate standard normal variable. Weng and Lin derived a corollary based on a lemma by Woodroofe (1989), which handles computing the expectation of a "nearly normal" distribution of the form $\phi(\mathrm{z}) f(\mathrm{u}|\mathrm{z})$. Using the corollary, if the posterior distribution of z can be expressed in the form of Eq. 4, then the first and second moment of the posterior distribution of z, assuming $f(\mathrm{u}|\mathrm{z})$ is twice differentiable, can be found as

$$E(z|\boldsymbol{u}) = E\left(\frac{\nabla f(u|z)}{f(u|z)}|u\right) \tag{5}$$

$$E\left(z_i^2|\boldsymbol{u}\right) = 1 + E\left(\frac{\nabla^2 f(\boldsymbol{u}|z)}{f(\boldsymbol{u}|z)}\middle|\boldsymbol{u}\right), i = 1, \ldots, p \tag{6}$$

where, by definition, the ith component of $\nabla f(\mathrm{u}|\mathrm{z})$ is $\frac{\partial f(\boldsymbol{u}|z)}{\partial z_i}$ and the $(i,j)th$ entry of $\nabla^2 f(\mathrm{u}|z_i)$ is $\frac{\partial^2 f(\boldsymbol{u}|z)}{\partial Z_i Z_j}$. Equations 5 and

6 are the key conclusions from the corollary that form the basis of the follow-up derivations.

The first moment can be further simplified by applying the chain rule to the first derivative in Eq. 5. For a single parameter z_i, after observing a response u_{ij},

$$E\left(z_i|u_{ij}\right) = E\left(\frac{\frac{\partial f\left(u_{ij}|z\right)}{\partial z_i}}{f\left(u_{ij}|z\right)}\middle|u_{ij}\right) = E\left(\frac{\partial \log f\left(u_{ij}|z\right)}{\partial z_i}\middle|u_{ij}\right) \tag{7}$$

The variance can be found as

$$\begin{aligned}\mathrm{var}\left(z_i|\boldsymbol{u}_{ij}\right) &= E\left(z_i^2|u_{ij}\right) - E\left(z_i|u_{ij}\right)^2 \\ &= 1 + E\left(\frac{\nabla^2 f\left(u_{ij}|z\right)}{f\left(u_{ij}|z\right)}\middle|u_{ij}\right) - E\left(\frac{\partial \log f\left(u_{ij}|z\right)}{\partial z_i}\middle|u_{ij}\right)^2\end{aligned} \tag{8}$$

Now, based on Eq. 3, the posterior means and variance of ξ_i are

$$\begin{aligned}\mu_i' &= E\left(\xi_i|u_{ij}\right) = \mu_i + \sigma_i E\left(z_i|u_{ij}\right) \\ &= \mu_i + \sigma_i E\left(\frac{\partial \log f\left(u_{ij}|z\right)}{\partial z_i}\middle|u_{ij}\right),\end{aligned} \tag{9}$$

$$\begin{aligned}\sigma_i'^2 &= \mathrm{var}\left(\xi_i|u_{ij}\right) = \sigma_i^2 \mathrm{var}\left(z_i|u_{ij}\right) \\ &= \sigma_i^2\left(1 + E\left(\frac{\nabla^2 f\left(u_{ij}|z_i\right)}{f\left(u_{ij}|z_i\right)}\middle|\boldsymbol{u}\right) - E\left(\frac{\partial \log f\left(u_{ij}|z\right)}{\partial z_i}\middle|u_{ij}\right)^2\right),\end{aligned} \tag{10}$$

which serve as the parameter estimate update formula. indeterminacy inherent in IRT can be resolved by setting $\mu_{\theta_i}^{(0)} = 0$ and $\sigma_{\theta_i}^{2\,(0)} = 1$. Here, in this method, the priors serve as initial values. After responding to m items, the current parameter estimates become $\mu_{\theta_i}^{(m)}$ and $\sigma_{\theta_i}^{2\,(m)}$. Suppose the response to the $(m+1)th$ item is correct, then the next parameter estimates are given as

$$\mu_{\theta_i}^{(m+1)} = \mu_{\theta_i}^{(m)} + \frac{\mu_{\beta_j}^{(m)}\sigma_{\theta_i}^{2\,(m)}}{\nu^{(m)}}\frac{\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} \tag{11}$$

$$\sigma_{\theta_i}^{2\,(m+1)} = \sigma_{\theta_i}^{2\,(m)}\max\left(1 - \left(\frac{\mu_{\beta_j}^{(m)}\sigma_{\theta_i}^{(m)}}{\nu^{(m)}}\right)^2\left[\frac{x^{*(m)}\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\Phi\left(x^{*(m)}\right)^2}\right], \kappa\right) \tag{12}$$

where $x^{*(m)} = \frac{\mu_{\alpha_j}^{(m)} + \mu_{\beta_j}^{(m)}\mu_{\theta_i}^{(m)}}{\nu^{(m)}}$, $\nu = \sqrt{1 + \sigma_{\alpha_j}^{2\,(m)} + \sigma_{\beta_j}^{2\,(m)}\mu_{\theta_i}^{2\,(m)} + \sigma_{\theta_i}^{2\,(m)}\mu_{\beta_j}^{2\,(m)}}$. If the $(m+1)th$ response is incorrect, the equations are

$$\mu_{\theta_i}^{(m+1)} = \mu_{\theta_i}^{(m)} - \frac{\mu_{\beta_j}^{(m)}\sigma_{\theta_i}^{2\,(m)}}{\nu^{(m)}}\frac{\phi\left(x^{*(m)}\right)}{1 - \Phi\left(x^{*(m)}\right)} \tag{13}$$

$$\sigma_{\theta_i}^{2\,(m+1)} = \sigma_{\theta_i}^{2\,(m)}\max\left(1 - \left(\frac{\mu_{\beta_j}^{(m)}\sigma_{\theta_i}^{(m)}}{\nu^{(m)}}\right)^2\left[\frac{-x^{*(m)}\phi\left(x^{*(m)}\right)}{1 - \Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\left[1 - \Phi\left(x^{*(m)}\right)\right]^2}\right], \kappa\right) \tag{14}$$

Although Weng and Coad (2011) applied the above results to a graded response model (GRM), we will consider a two-parameter normal ogive (2PNO) model in this study, which is a special case of GRM. Assuming person and item parameters follow normal priors and are mutually independent, it is possible to derive a series of equations to update the means and variance that define their posterior distributions. Meanwhile, setting a prior for the underlying distribution for examinees solves the scale indeterminacy issue for the item calibration (Kim, 2006). The primary challenge here is to evaluate the expectations on the right-hand sides of Eqs. 9 and 10, and some sort of approximation is necessary for non-iterative update rules that are simple to compute. In the derivation given in Weng and Coad (2011), the most important idea is to replace the random variables with their corresponding means.

To be specific, let prior means and variance of the ability parameter be $\mu_{\theta_i}^{(0)}$ and $\sigma_{\theta_i}^{2\,(0)}$, respectively. Note that the scale κ is a constant slightly larger than 0 to make sure that posterior variance is always positive. We used $\kappa = 0.001$ in the study. The derivations that are originally provided by Weng and Coad (2018).

Based on Eqs. 11 and 13, it is evident that the posterior mean of the ability parameter goes up after a correct response (plus an always positive value since β_j is always positive and Φ is a CDF) and goes down after an incorrect response, which makes intuitive sense and also agrees with the ERS formulas. Another noteworthy observation from the equations is that the next updated estimate only depends on the response data and the parameter estimates from the previous cycle. This is analogous to the "memoryless" property of the Markov chains and means that for the purpose of tracking learner ability, it is possible in practice to only extract the most current state of the system rather than the learner's entire response history. This contrasts with CAT and online calibration in which learner ability has to be estimated using the response vector and parameters of all the previously answered (operational) items.

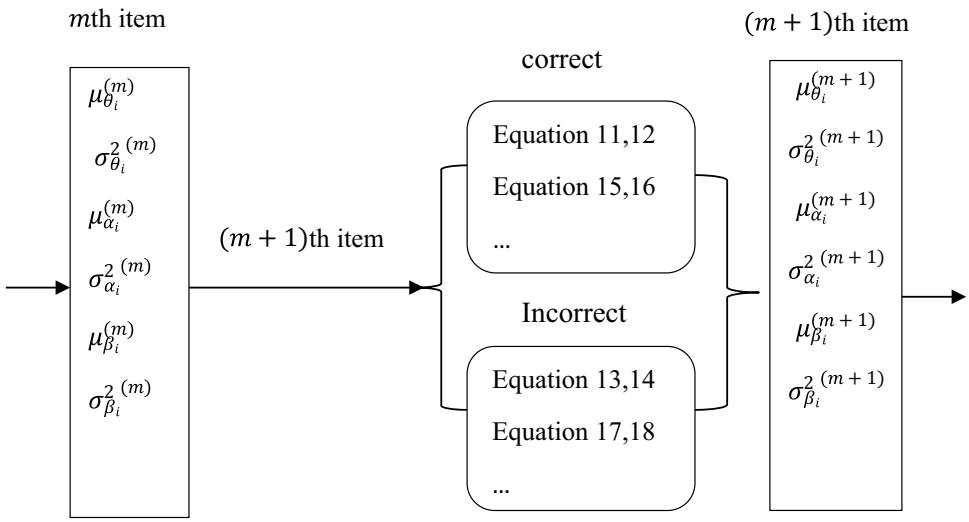


Fig. 1 Flowchart of parameter update using the Bayesian algorithm

The update formulas of the item parameters for binary models can be similarly obtained. Take the intercept parameter as an example. For the correct and incorrect response, the intercept parameter can be updated as follows:

$$\mu_{\alpha_i}^{(m+1)} = \mu_{\alpha_i}^{(m)} + \frac{\sigma_{\alpha_j}^{2\,(m)}}{v^{(m)}} \frac{\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} \tag{15}$$

$$\sigma_{\alpha_i}^{2\,(m+1)} = \sigma_{\alpha_i}^{2\,(m)} \max\left(1 - \left(\frac{\sigma_{\alpha_j}^{(m)}}{v^{(m)}}\right)^2 \left[\frac{x^{*(m)}\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\Phi\left(x^{*(m)}\right)^2}\right], \kappa\right) \tag{16}$$

$$\mu_{\alpha_i}^{(m+1)} = \mu_{\alpha_i}^{(m)} - \frac{\sigma_{\alpha_j}^{2\,(m)}}{v^{(m)}} \frac{\phi\left(x^{*(m)}\right)}{1 - \Phi\left(x^{*(m)}\right)} \tag{17}$$

$$\sigma_{\alpha_i}^{2\,(m+1)} = \sigma_{\alpha_i}^{2\,(m)} \max\left(1 - \left(\frac{\sigma_{\alpha_j}^{(m)}}{v^{(m)}}\right)^2 \left[\frac{-x^{*(m)}\phi\left(x^{*(m)}\right)}{1 - \Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\left[1 - \Phi\left(x^{*(m)}\right)\right]^2}\right], \kappa\right) \tag{18}$$

Here the intercept also increases with a correct response, but because the intercept is in the opposite direction of item difficulty ($\alpha_j = -a_j b_j$), the difficulty parameter actually drops given a correct response, which intuitively means the item is not as difficult as previously thought and is again consistent with the ERS formulas.

The process of parameter update is further illustrated by Fig. 1, which also highlights the "memoryless" property of the new update rules. Parameter estimates are performed each time a learner responds to an item. After the response is recorded and scored, only the most current parameter estimates of the learner and the item are fed into a set of equations, depending on whether the response was correct, from which the updated parameter estimates are to be calculated for both the learner and the item. These new parameter estimates replace the previous ones and will be used in the next update when needed, and the process continues until the entire session is finished.

Some special considerations are outlined when implementing the Bayesian update rules in adaptive learning system. Firstly, the order of data entry is a unique issue because item and person parameters are now computed at the same time, thus affecting each other. The current study follows a "by-person" design assuming a user starts the session after the previous user has finished. The reason is that for online learning platforms, as well as online surveys, people may enter the system at varying entry points, and they may interact with the system at their own pace. The procedure is illustrated in Fig. 2. Learner i completes L items, and by the time he/she is finished, the next learner starts his/her own set of items. Note that here the item numbering simply refers to the item position on each person's

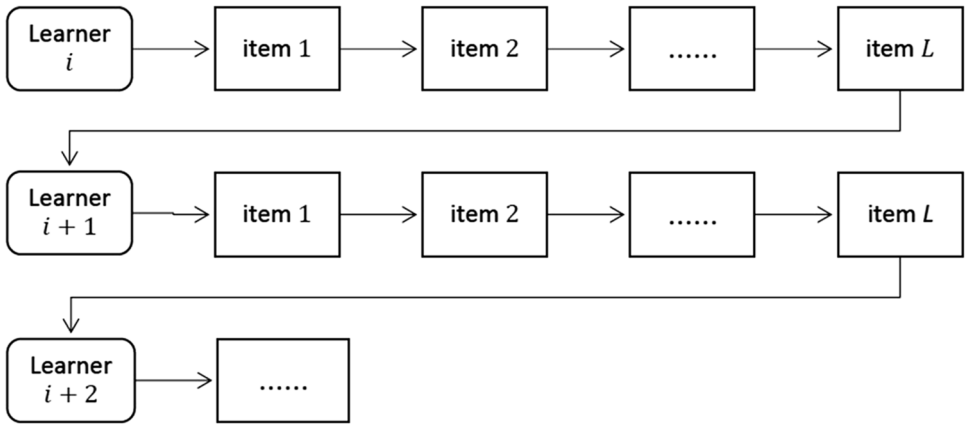


Fig. 2 By-person design procedure

test, and an item j may not be the same item for different examinees in adaptive testing. This design might put the early cohort at a disadvantage as they may have larger posterior variance, regardless of their ability levels, in part because the items given to them are barely calibrated and have large posterior variance as well. However, this becomes less of an issue in the learning environment, since the order is not identical for all sessions and the ordering effect will be canceled out over multiple sessions. In a real online testing scenario, it will not be strictly sequential "by person," but we use this "by-person" design as a simplified imitation in our simulation study because otherwise we also need to simulate detailed time stamps for all item-person encounters, which complicates the design unnecessarily. However, the Bayesian update method can be applied in any scenario because it is designed to update both item and person parameters whenever a response is recorded by the system.

Secondly, the Bayesian update algorithm alleviates the scale indeterminacy by fixing the prior mean and prior variance of θ. However, this constraint is no longer sufficient in adaptive learning system when θ changes due to learning. Instead, the scale could be fixed using anchor items whose parameters are pre-calibrated (Wang et al., 2016). The anchor items can either be a common set of linking items that appear in all sessions or completely different across sessions. In this study, the latter is preferred because it is better to avoid administering the same items repeatedly to avoid potential learning effect.

Maximum posterior weighted information (MPWI)

In an adaptive learning system, the item assignment rule plays an important role in selecting the appropriate next item for each learner. We propose a modified version of the maximum posterior weighted information (MPWI), which is a Bayesian item selection criterion that integrates over the posterior distribution of learner ability (van der Linden, 1998). Because in the Bayesian update rule, the new item parameters were also described by their posterior distributions, the MPWI is modified to integrate over the item parameter distributions and it is evaluated using Monte Carlo integration as follows: further assuming item parameters and θ are also independent of each other,

$$MPWI = \frac{1}{D}\sum_{d} I\left(\theta_{id}, \alpha_{jd}, \beta_{jd}\right) \tag{19}$$

where θ_{id}, α_{jd},β_{jd} are the dth draw from the correspondin posterior distributions with the total number of draws D set to 500, $I(\theta_i, \alpha_j, \beta_j)$ denotes the Fisher information.

In previous simulation studies that involve calibrating items with discrimination parameters (Makransky & Glas, 2010; Ren et al., 2017), the calibration processes were severely affected by uneven item exposure. Some items were rarely chosen by the selection criteria and stayed in the calibration phase for an extended period of time before being put to operational use. The restrictive progressive method, an item exposure control method proposed by Wang et al. (2011), is used to balance the frequencies of item delivery. Combining it with the MPWI, the final selection criterion is written as

$$\left(1 - \frac{r_j}{r_{\max}}\right)\left[\left(1 - \frac{y}{L}\right) R_j + MPWI_j^* \times \frac{by}{L}\right] \tag{20}$$

where y is the number of items already administered, and r_j is the exposure rate of item j. $r_{\max}$ is the prespecified maximal exposure rate and was set to 0.40 in this study. R_j denotes the random component and was sampled from a uniform distribution between 0 and the maximum of $MPWI$ among all candidate items. b is a weighting coefficient that balances the trade-off between exposure control and estimation accuracy, with values larger than 1 indicating that estimation accuracy is prioritized. In this study it was set to 2, following Wang et al. (2011).

The item selection indices are calculated for each eligible item remaining in the pool. For pre-calibrated items the true item parameters are used such that MPWI is computed by only integrating over θ, while for the new item the indices are computed from the current parameter estimates. Because the MPWI method considers the slope/discrimination parameter, it is expected to result in better ability parameter recovery compared with the random selection or matched-difficulty (MD) method.

Simulation study

A simulation study was conducted to evaluate the performance of the Bayesian update algorithm and MPWI item assignment criterion as a usable and useful technique in the computerized adaptive learning environment. The simulations and procedures were performed in R 3.6.1 (R Core Team, 2019).

Design

We designed an online learning setting that includes multiple sessions between which changes of learner ability occur. The new algorithm was tasked with calibrating item parameters and tracking ability changes at the same time, and its performance was evaluated against a modified version of online calibration as a baseline.

Data generation A total of four sessions were simulated, with each session having 20 or 30 items, and 3000 θs were randomly sampled from the standard normal distribution. That is, each of the 3000 students completed 80 or 120 items in four different sessions, with an interval between each of them. A real-world example would be a student taking a weekly online course with assignment for four consecutive weeks. Following Weng and Coad (2018), the intercept and ability parameters were generated from $\alpha_j \sim \mathrm{N}(0, 1^2)$ and $\theta_i \sim \mathrm{N}(0, 1^2)$, respectively. These distributions also served as the priors. In their simulation, they generated β_j from $\mathrm{N}(1, 20)$ because they were evaluating the performance of the algorithm for online rating data, which, unlike in educational settings, allowed for negative discrimination. In this study the slope parameter was instead generated from $\beta_j \sim \mathrm{N}(1, 0.32^2)$ and the prior distribution was $\beta_j \sim \mathrm{N}(1, 0.6^2)$. Note that priors for the Bayesian update rule also refer to the initial values, which means that all the new items have the same intercept and slope parameters at the beginning. Responses were generated from the two-parameter binary normal ogive model using true item and ability parameters, and a set of parameter estimates was obtained with the Bayesian update rules after every new response.

The item bank consisted of 500 items whose parameters were assumed to be constant across sessions, and all sessions shared a common item bank. The bank size was relatively small by CAT standard as with 30-item sessions the ratio of test length to bank was close to 1:4, but this is offset by the fact that in learning environments, items are allowed to be delivered repeatedly. However, to keep learners from memorizing the item, the same item could not appear in the same session, and an item could appear in no more than two sessions.

The change pattern of ability parameter was assumed to only involve between-session change and simulated as follows. Ability was assumed to be constant within each session, and after each session, a change parameter randomly sampled from $N(0.2, .0.1^2)$ was added to each ability parameter. As a result, each individual would have a different growth pattern, and a drop in ability is possible in some cases. Because there usually is a time interval between sessions, the uncertainty about the ability estimate would be increased during this time period. To account for the increased variability, a value was added to the posterior standard deviation (SD) of each ability parameter at the beginning of each new session except the first. Suppose at the end of a session the posterior SD of ability of learner i is updated as $\sigma_{\theta i}$, then at the start of the next session, the posterior SD is set to be $\sigma_{\theta_i}' = \sigma_{\theta i} + \tau$ where τ is the added SD. A similar approach was taken in previous research where the τ was either chosen a priori or estimated from the first batch of data (Glickman, 1999; Weng & Coad, 2018). In this simulation τ was chosen to be 0.5, and pilot simulations had found that any number between 0.25 to 0.75 would yield similar and reasonable results. Note that the added value only applied to the ability variance, while item parameter and the means of ability did not need special adjustment when transitioning between sessions thanks to the addition of pre-calibrated items, which would be explained next.

Pre-calibrated items As we mentioned earlier, changing abilities across sessions requires pre-calibrated items to establish a common scale. The question remains with regard to the proportion of items in the bank that should be pre-calibrated, as it is ideal to have as many uncalibrated items as possible to save the cost of item calibration. Based on the recommendation that at least 20% of the items should be used for linking purpose (Kolen & Brennan, 2014), the lower bound of the proportion of pre-calibrated items was set to 20%. Another level at 50% was chosen as an upper bound to determine whether having more known items in the bank would boost the measurement performance of the algorithm. If the system could still function reasonably well with as low as 20% of items being pre-calibrated, it would greatly reduce the cost of item bank construction. In this case, only a small

number of starting items are needed while the majority of the items can still be calibrated on the fly.

Item selection For the adaptive delivery of items, three item selection methods were considered. Random selection was used as a baseline condition and expected to yield the worst results for ability estimation. MD was used because it represented the item selection methods for ERS/Rasch models. It simply selected the candidate item with the smallest absolute difference between the current ability level and the item difficulty. As with MPWI, MD was restricted to a maximum item exposure rate of 0.4. The third item selection method is restricted progressive MPWI.

Comparison with online calibration The Bayesian update rule also requires having pre-calibrated items at the beginning, which makes the setting comparable to that of online calibration, although in online calibration studies, the new items are not truly estimated on-the-fly, but rather calibrated offline after data collection concludes. Hence, in online calibration setting, the new items were randomly assigned. To see whether the Bayesian update rule together with MPWI could offer some improvement over the existing methods, online calibration was added as a fourth condition in addition to the three item selection methods. The operational items from online calibration were simply the pre-calibrated items discussed above, with their proportion also being varied at 20% and 50%.

Unlike random selection, in online calibration the parameters were estimated only once after each session[1]. There was no need to obtain interim parameter estimates because no adaptive item selection or ability estimation took place in the middle of a session. After the end of a session, new item parameters were first computed by marginal maximum likelihood estimation (MMLE) with multiple expectation-maximization cycles (MEM) with the pre-calibrated item parameters fixed (Ban et al., 2001), and then ability parameter was found by EAP using the estimated new items and the pre-calibrated items. To account for the changing abilities, the population means and variances were allowed to vary across sessions and estimated during the MEM process (Wang et al., 2020). Note that in the original online calibration design, only operational items are used for scoring. This was not feasible in the current simulation because in some conditions there were very few operational items in a session. In addition, using all items for scoring makes the online calibration estimation more comparable to the Bayesian update rules which also score learners based on all items. The estimated new item and ability parameters were then used as the starting values of the next session, and the process was repeated until all four sessions were finished. For fair comparison, the initial parameters for all new items were the same for all methods.

Evaluation criterion In summary, there were a total of 2 levels of session length (20/30 items per session) ×2 levels of pre-calibrated items (20%/50%) × 4 levels of item selection method/design (random/MD/MPWI/online calibration) =16 conditions. In each condition, the same item bank was used to generate responses in 30 different replications. The bias and root-mean-square error (RMSE) of ability estimate were reported by session, while for item parameter recovery both the by-session and the final estimates were presented. In addition, the posterior means and SD of ability estimates after every update were recorded for each learner, from which a trace plot showing the trajectory of mean and SD estimates during the sessions could be generated at the individual level. Item exposure was examined by calculating the exposure rate r_j (number of administrations divided by the sample size) for every item in the bank within each replication. In addition, the standard deviation of the exposure rate was computed per replication to demonstrate the balance of exposure rate (Wang et al., 2014). Both mean and standard deviation of item exposure were reported separately for pre-calibrated and new items. These results were averaged across the 30 replications and were compared for three item selection methods.

Results

Item calibration

The recovery measures of item parameters from the two session lengths are summarized in Table 1. The 30-item sessions clearly outperformed the 20-item sessions in all conditions in terms of both indices, with the largest discrepancy coming from the online calibration condition. In both session lengths, MD had the smallest bias for the slope parameters, while for intercept parameters MD and online calibration had similar bias that were both smaller than those of random selection and MPWI under the 30-item session with 50% pre-calibrated items. There was also a tendency for the Bayesian update rules to overestimate the intercept parameter and underestimate the slope parameter, while both parameters were overestimated in online calibration that utilized MMLE/EM.

RMSE showed a different pattern in which random selection was clearly superior with all values below 0.20, followed by MPWI whose values ranged from 0.18 to 0.24.

[1] In the random selection condition, we still used the Bayesian update rule for estimating both item and person parameters, although these interim estimates were not actively used during item selection. Hence, the comparison between the random and online calibration conditions merely reflects the performance difference of the Bayesian update rule and MEM as two parameter estimation methods.

Table 1 Parameter recovery of new items

Session length	Item selection/Design	Pre-calibrated items	Bias		RMSE	
			α	β	α	β
20	Random	20%	0.090	−0.118	0.149	0.169
		50%	0.083	−0.106	0.143	0.163
	Matched-difficulty	20%	0.068	−0.045	0.236	0.309
		50%	0.043	−0.030	0.224	0.297
	MPWI	20%	0.064	−0.103	0.218	0.235
		50%	0.049	−0.092	0.212	0.239
	Online calibration	20%	0.079	0.130	0.370	0.364
		50%	0.041	0.125	0.312	0.355
30	Random	20%	0.072	−0.094	0.122	0.137
		50%	0.067	−0.089	0.118	0.133
	Matched-difficulty	20%	0.050	−0.036	0.209	0.266
		50%	0.031	−0.032	0.202	0.249
	MPWI	20%	0.060	−0.064	0.188	0.196
		50%	0.044	−0.056	0.183	0.193
	Online calibration	20%	0.052	0.080	0.253	0.260
		50%	0.032	0.084	0.219	0.258

Although having the smallest bias, MD yielded larger RMSE than random selection and MPWI, especially for the slope parameters. Online calibration performed the worst overall with all RMSE values approaching 0.31 for 20-item session conditions.

The proportion of pre-calibrated items showed a small yet consistent effect on item parameter recovery. As expected, having more pre-calibrated items resulted in better recovery of the new items However, the effect was only modest in many conditions, in which adding 150 pre-calibrated items (there were 100 pre-calibrated items in the bank in the 20% condition and 250 in the 50% condition) only reduced the RMSE by about 0.10. It could be said that with only 20% of items being known, the Bayesian update algorithm still functioned well.

The bias and RMSE of item parameter estimate over the four sessions are presented in Figs. 3, 4, 5, and 6. Three item selection methods combining with Bayesian update rules showed an ascending trend of bias over four sessions, whereas the online calibration algorithm almost stayed the same across sessions. One potential reason is that the θ distribution matches well with the item difficulty (i.e., both θ and b are from N(0, 1)) in the first session, whereas in the later sessions, the mean of θ is higher than average difficulty (due to learning), and therefore the mismatch yields larger bias for item parameter estimate. Such an ascending trend of bias due to mismatch shows up for all methods except online calibration (OC), because in OC, parameter estimates from a preceding session is used as initial values in the next session, which helps combat bias. That said, when looking at the RMSE for slope estimates (see Fig. 6), we observe a descending trend except the OC method. That is because in OC, the parameters are estimated from response data collected in each session separately, resulting in smaller sample size and larger standard error. Even though the bias is stable for OC across four sessions, the large standard error still makes RMSE from OC the highest.

Ability estimation

Bias The bias of ability estimates over the four sessions is displayed in Fig. 7. The Bayesian algorithm systematically underestimated learners' ability across most conditions, except for the first session after which all item selection method/designs had minimal bias. It was also obvious from the figures that bias tended to accumulate as learners progressed through later sessions except in the random condition. As a result, random selection had the least bias overall with any of the (absolute) values smaller than 0.05. The trend was somewhat mitigated by the higher percentage of pre-calibrated items, but not by the increased session length. Among the three remaining item selection design/methods, MPWI performed better than the two adaptive selection methods (MD and online calibration) in all conditions except for the 30-item with 50% pre-calibrated items condition. Under this condition, it is slightly worse than online calibration.

RMSE The RMSE of ability estimates was visualized in a similar fashion in Fig. 8. Like bias, RMSE in all conditions showed an overall increasing trend over time. The trend was also flattened, although far from completely, by the higher

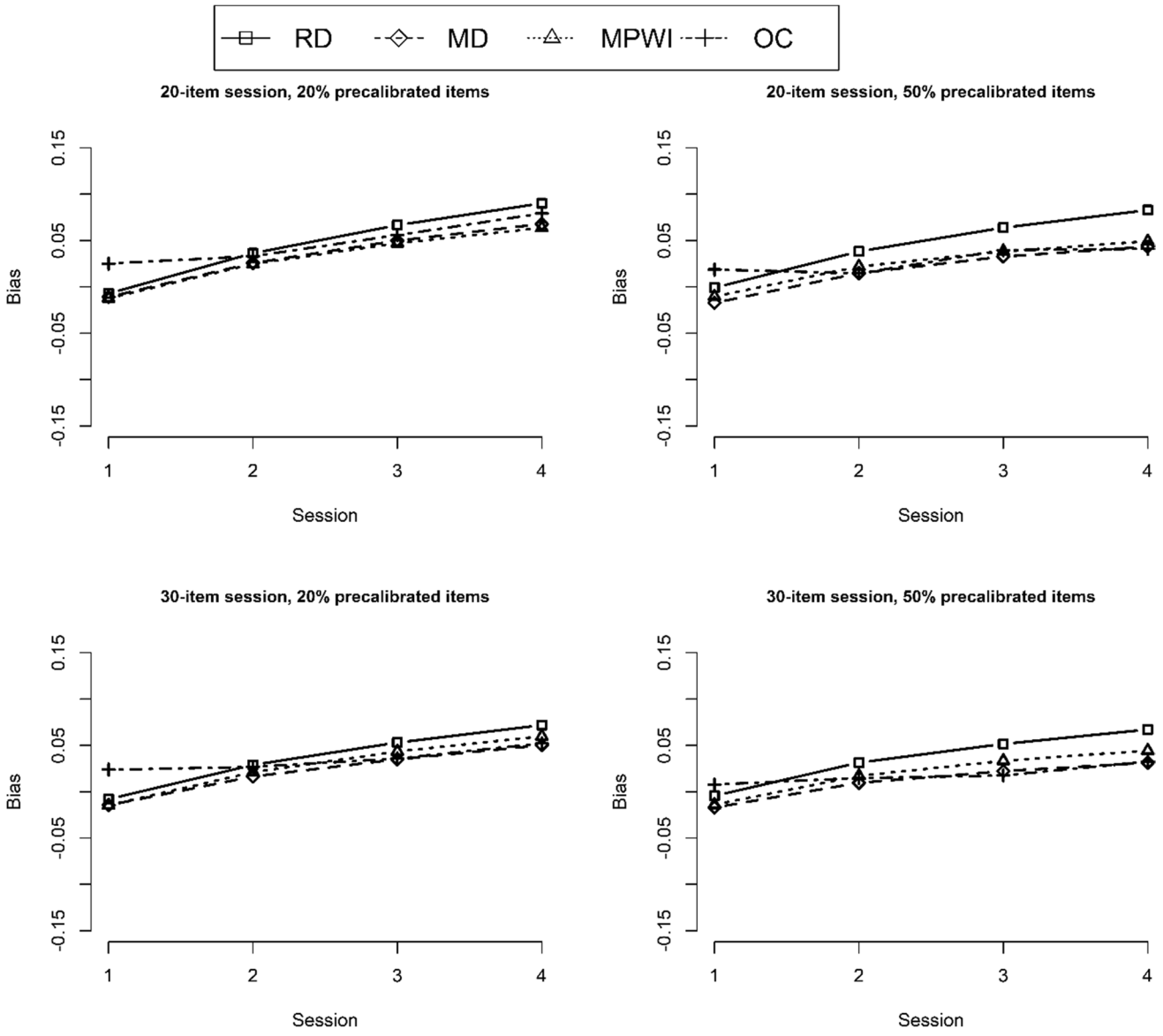


Fig. 3 Bias of intercept estimates over four sessions

proportion of pre-calibrated items, but persisted in the face of increased session length. However, unlike bias, the longer session did bring down the RMSE values consistently by at least 0.05 across all conditions.

Random selection and online calibration showed results that were both considerably worse than their adaptive counterparts. The difference between random and adaptive designs was much larger in the 20% pre-calibrated condition, again thanks to the higher quality item parameter estimates. Results of MD and MPWI were almost identical for the first session, but in the remaining three sessions MPWI clearly outperformed the best by a margin of at least 0.02. The advantage of MPWI was bigger in less favorable conditions, such as shorter sessions or fewer pre-calibrated items. In the best-case scenario of 30-item session with half of the items known, MPWI was able to register RMSE values that ranged from 0.21 to 0.25.

Trace plot Figure 9a and b shows the estimates of the posterior means and SD, produced by the new Bayesian update algorithm, as a function of the items administered. The data came from two learners that were each randomly chosen from the simulation sample to illustrate the process of parameter update within and between sessions. The vertical dashed lines, like previous figures, marked the end/beginning of a session, with the data point at Item 0 denoting the starting value. Each mean and SD estimate, which was updated after every item answered, is represented by a solid

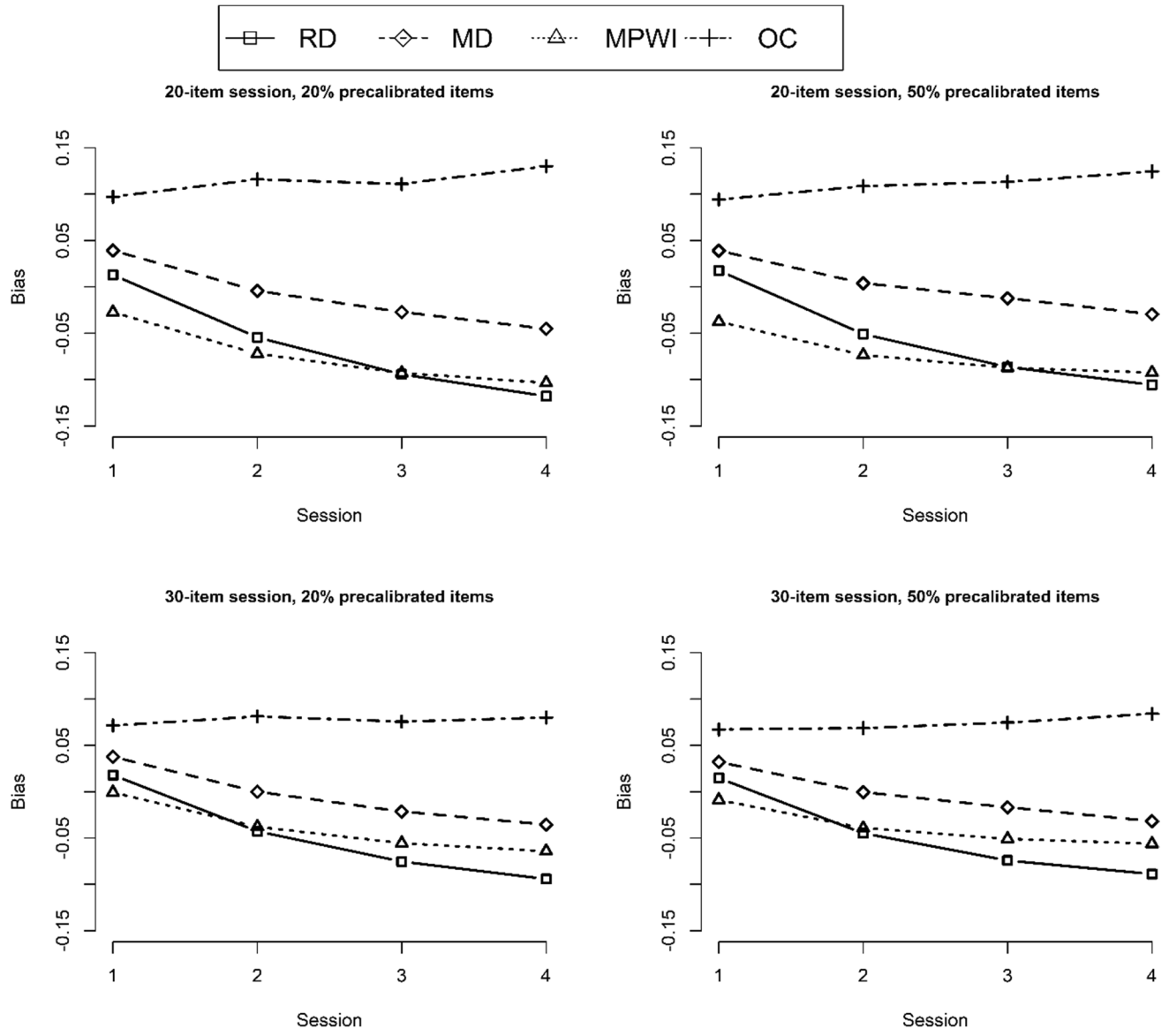


Fig. 4 Bias of slope estimates over four sessions

circle, and connected by lines to better depict the trend. The four dashed line segments in the upper figure denoted the true ability level of the learner at the given session and were horizontal since no within-session growth was assumed.

Figure 9a was based on the condition of 50% pre-calibrated items in 20-item sessions with the items adaptively selected by MPWI. The learner had a relatively high starting ability, and the amount of ability growth after the first and third session was close to the average level at around 0.2, while the second session saw only below average improvement. Starting at 0.0, since no other information of the learner was known (cf. Park et al., 2019b), the ability estimates had some major fluctuations before stabilizing around the true ability level. The process moved on to the next session with the final update from the previous session serving as the starting value of the next session, and a similar pattern of fluctuating and stabilizing was observed in the remaining three sessions.

The second figure shows the trajectory of the posterior SD values of the same learner. As was predicted by Eq. A13 in the Appendix, the values were monotonically decreasing within each session. This helped explain why in the upper figure the update size of ability estimates became smaller towards the end of a session, as the update size was positively associated with the posterior variance of ability parameter. Note that in the simulation design a constant of

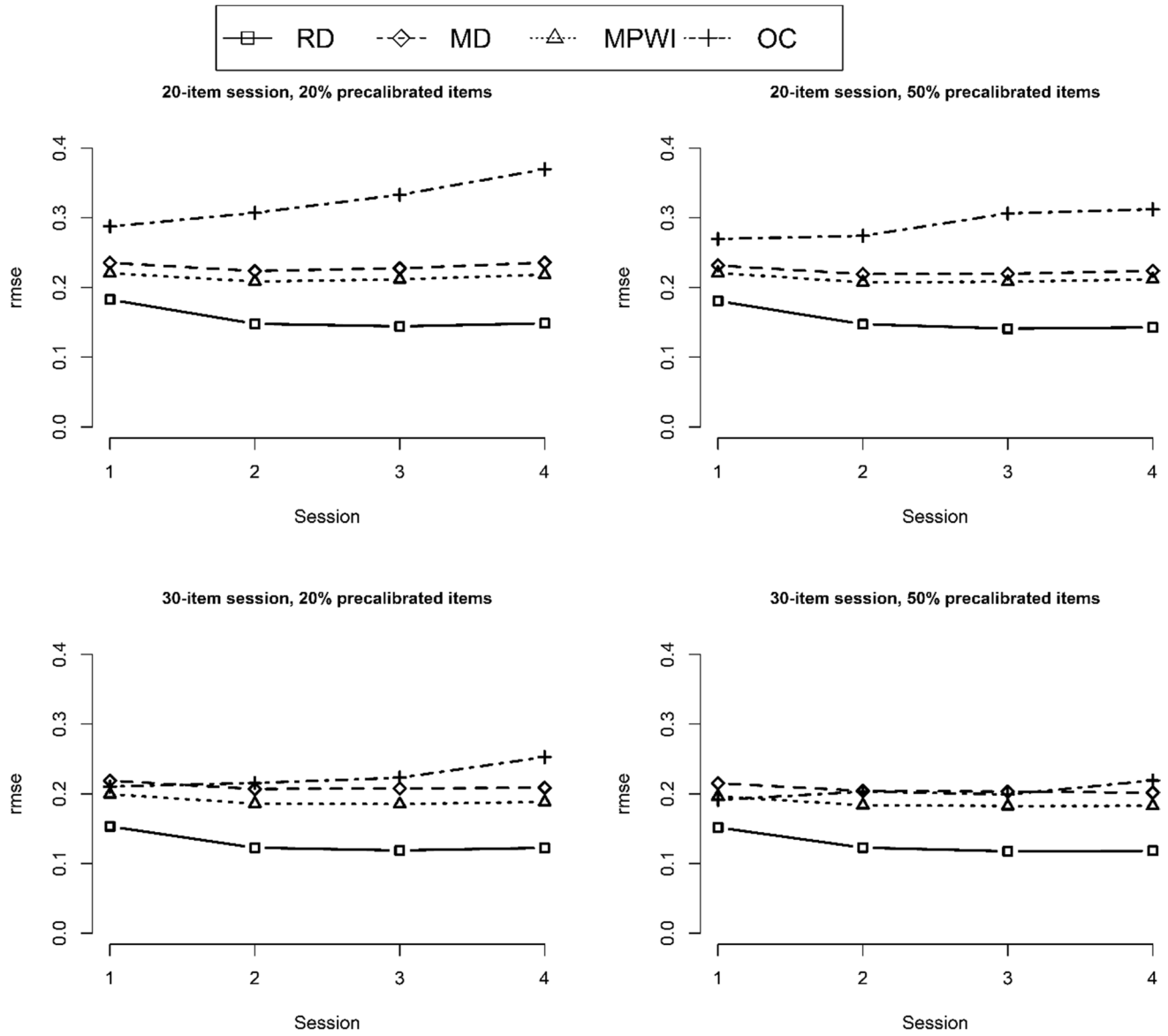


Fig. 5 RMSE of intercept estimates over four sessions

0.50 was added to the posterior SD at the beginning of a new session to account for the higher level of uncertainty. This was reflected by the abrupt increase of SD values between sessions.

In Fig. 9b, the data were taken from the 30-item sessions with all other settings being the same. The learner had a lower starting ability, and the ability estimates also had some substantial fluctuations in the first session. Even though the session was longer, the final estimate did not come as close to the true values as the previous example, probably because the system could not recover from the unexpected correct responses early in the sessions.

The SD plot at the bottom was almost identical to the 20-item sessions where within each session SD continuously decreased. Since the session became longer, the final posterior SD was also slightly smaller with values around 0.20. The 10 extra items also made it possible to see how the curve would gradually flatten out towards the end of a session; it is reasonable to expect that if the session continued indefinitely, the SD would still continue to drop until "converging" to a lower bound.

Item exposure

Table 2 showed the exposure statistics for the three item selection methods/designs, respectively. Results from the online calibration method were not presented here because it

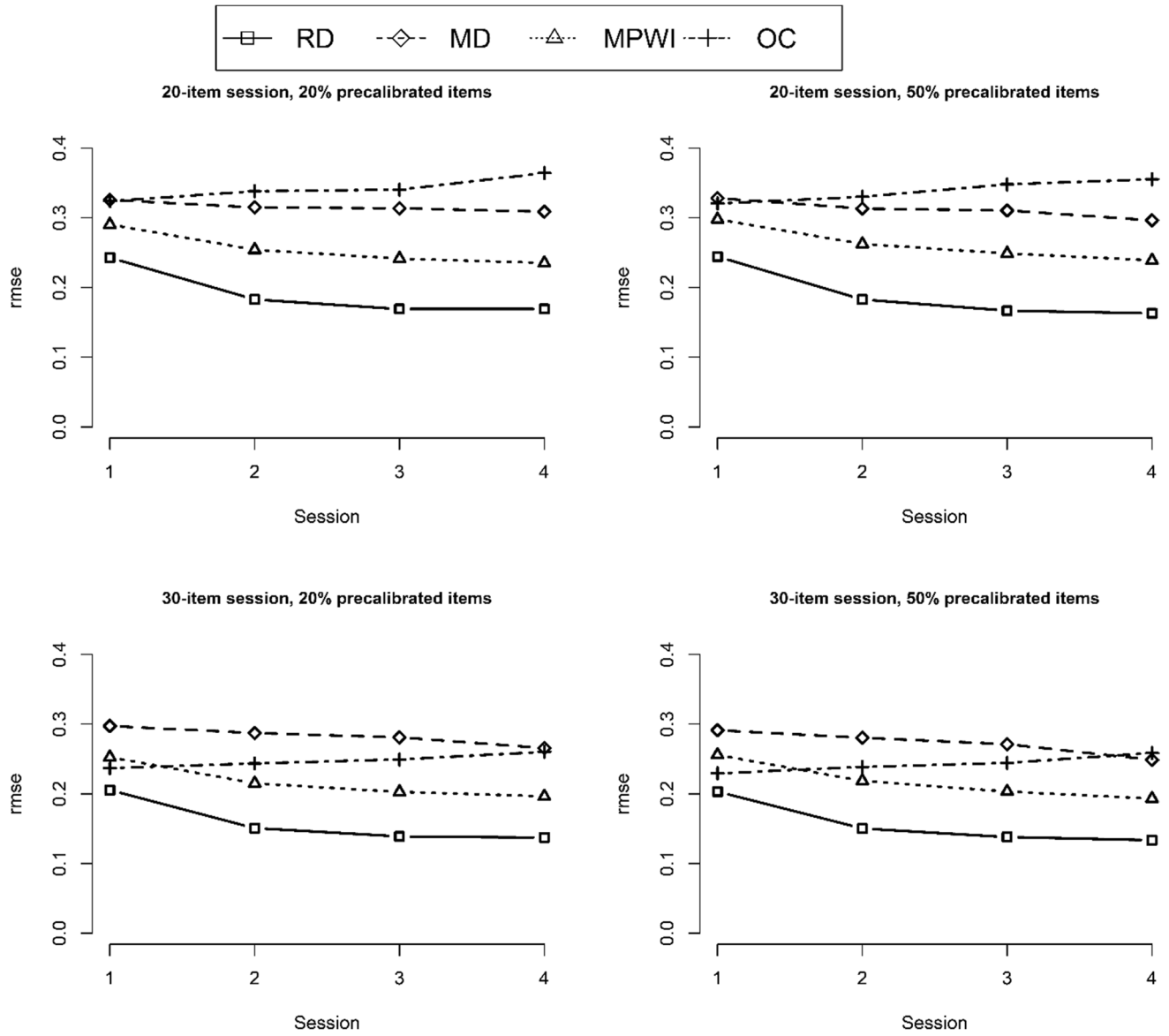


Fig. 6 RMSE of slope estimates over four sessions

used the same random selection. As expected, random selection had the most even item exposure rates, with very small standard deviation. There was no noteworthy difference between new and pre-calibrated items in terms of exposure rates. MD resulted in more unbalanced item administration, reflected by a much larger standard deviation of exposure rates. In addition, the average exposure rate of pre-calibrated items is slightly higher than that of the new items, implying that pre-calibrated items are more likely to be selected by the algorithm. With regard to MPWI, the variability in exposure rates further increased for both new and pre-calibrated items. The standard deviation was consistently larger in the 20-item condition; that is, item exposure was more balanced in the longer sessions. The disparity between pre-calibrated and new items showed a pattern similar to that observed in the MD condition. Generally, new items were relatively underexposed compared with the pre-calibrated items, but the difference was not very large, and new items still received an adequately large sample for calibration.

Discussion

An online Bayesian parameter estimation algorithm (Weng & Coad, 2018) was applied to an IRT-based adaptive learning environment, in which learners receive practice or

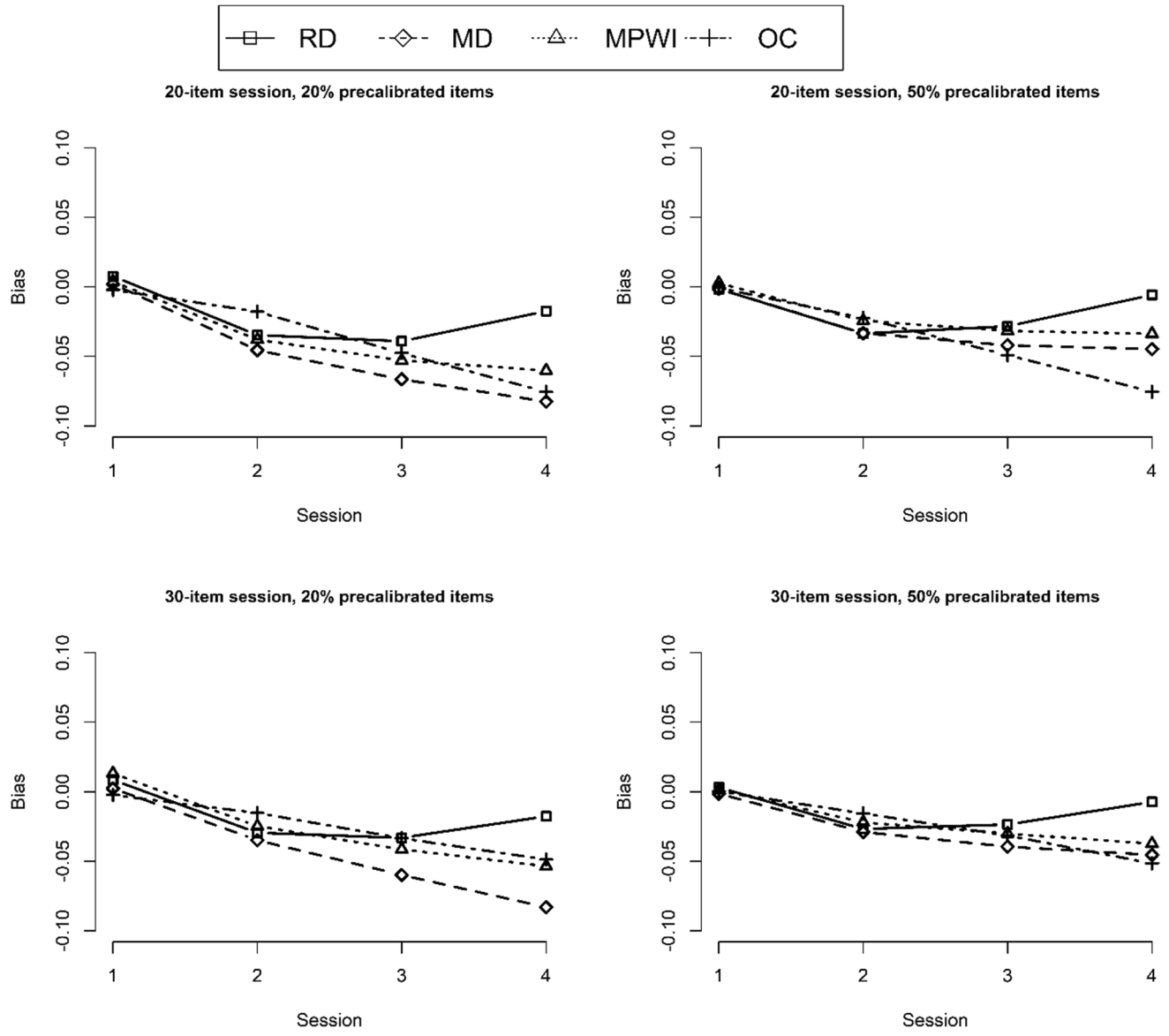


Fig. 7 Bias of ability estimates over four sessions

quiz items that are individually matched to their changing ability level. The performance of the Bayesian algorithm was explored through a multi-session measurement setting that mimicked a simplified online learning application. The goal was to evaluate the capability of the Bayesian algorithm to dynamically monitor learner performance and calibrate item parameters. Three factors were manipulated to explore how the algorithm would perform under different setup, and online calibration was added for comparison as a baseline. Results reveal that the Bayesian update algorithm, combined with modified MPWI item selection and exposure control measures, was able to achieve fast and reasonably accurate parameter estimations that were comparable to some of the existing psychometric methods. With exposure control measures implemented as recommended by Wauters et al. (2010), cases of extreme item exposure that occurred in some previous online calibration studies (Makransky & Glas, 2010; Ren et al., 2017) were not observed in the simulation, with only a few items being underexposed.

Regarding the three challenges of applying CAT procedures to the learning environment, the Bayesian update algorithm provides some promising alternatives to the topic of item bank calibration. By calibrating item parameters during item administration, online learning programs no longer need to obtain the parameters of all items before

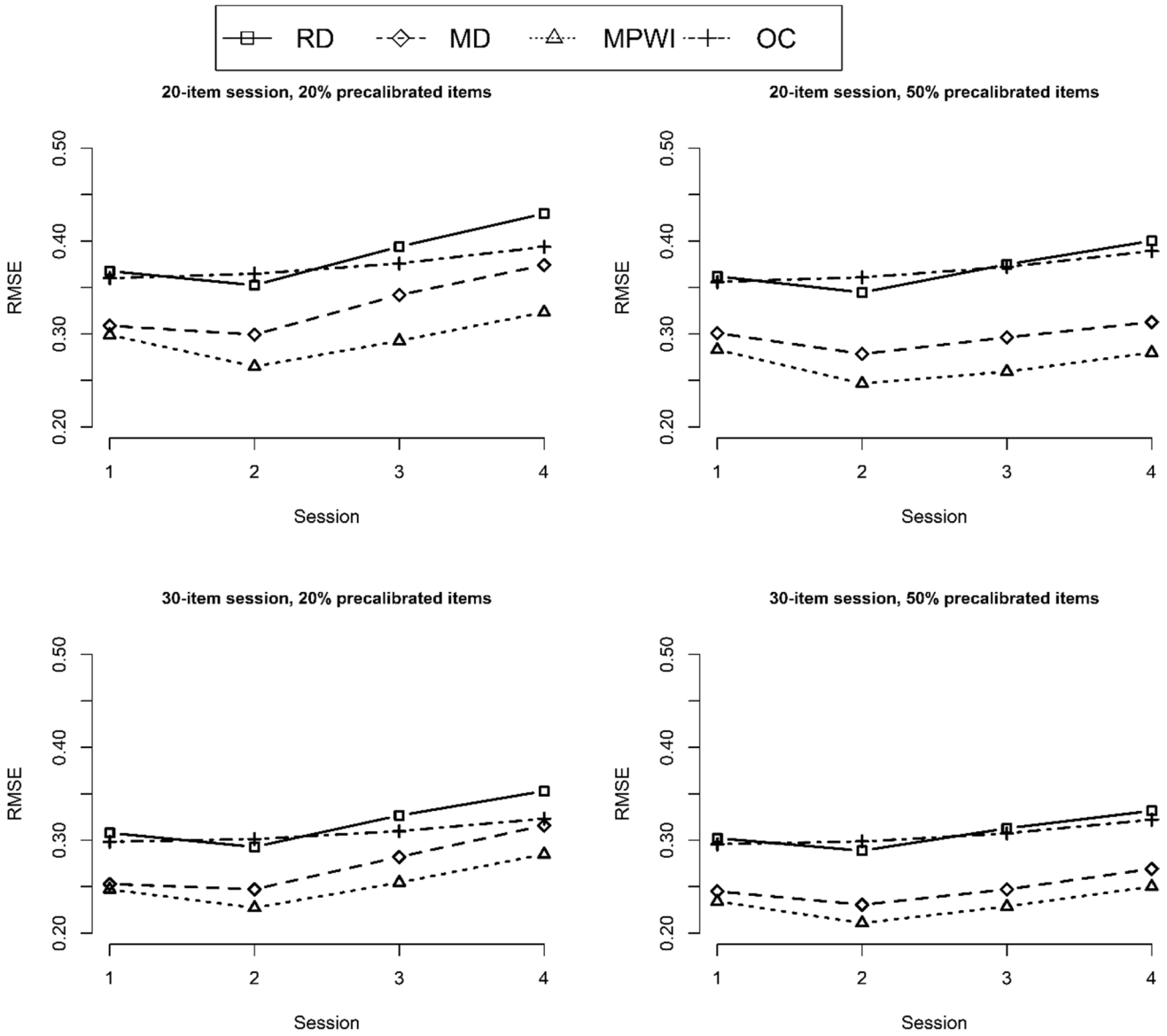
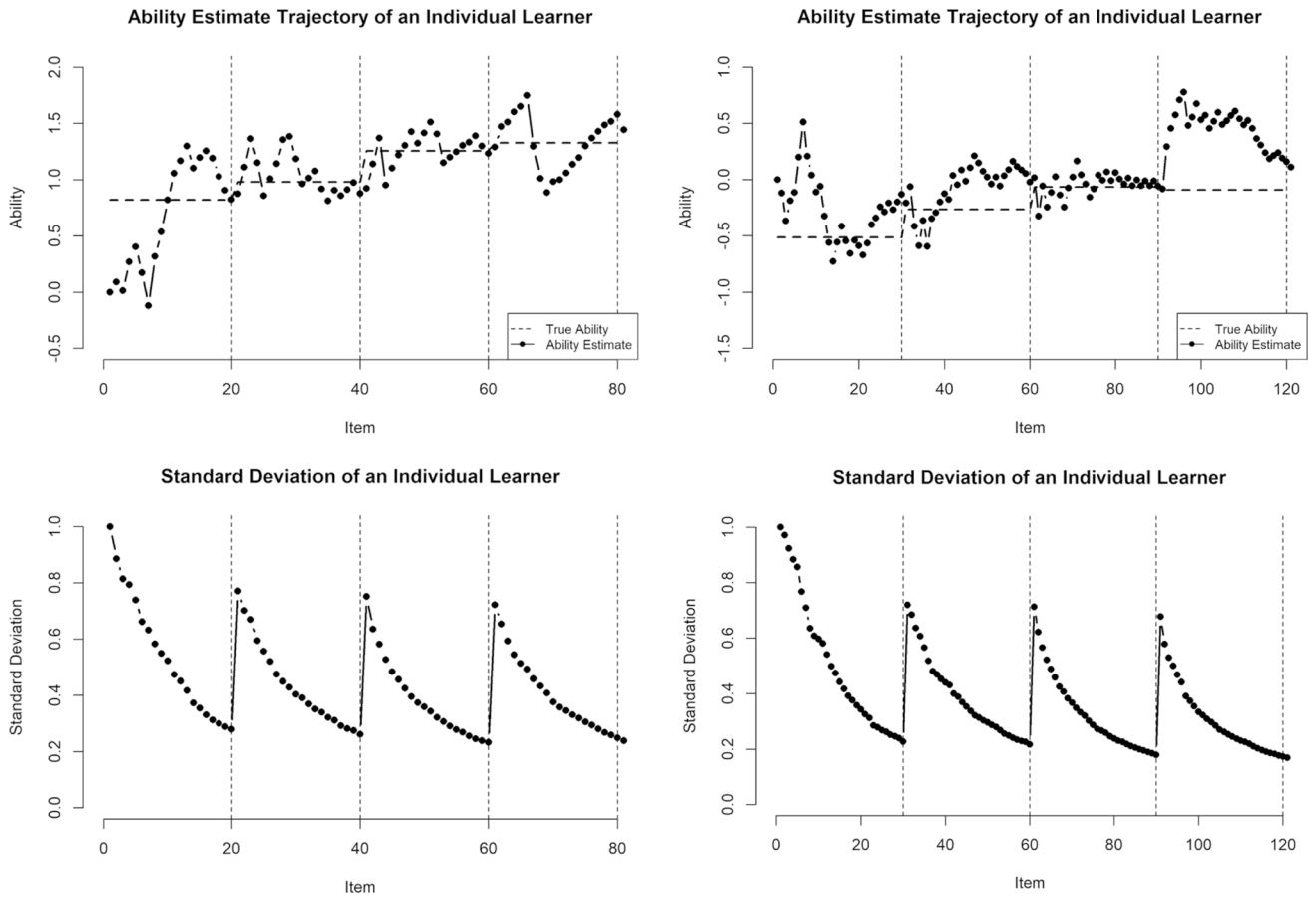


Fig. 8 RMSE of ability estimates over the four sessions

putting them to use. In the current simulation, with only 20% of items in the bank being pre-calibrated, the system was still able to function properly, recovering ability parameters and achieving adaptivity on the fly. This could help save a considerable amount of time and resources in the item bank construction phase, and since the new items could in turn qualify as pre-calibrated items once some threshold has been met (standard error or number of administrations, for instance), only a small number of starting operational items could snowball into a sizable, calibrated bank rather quickly.

The issue of dynamic ability tracking was also investigated in the simulations. Compared with similar research based on ERS, the Bayesian algorithm enjoys the noniterative, closed-form update property of ERS but offers a more sophisticated alternative that is more closely related to the traditional parameter estimation techniques used in IRT. In ERS, the key component of the algorithm, the update size K, is either fixed or calculated heuristically. The new update rules instead calculate the update size by approximating the parameter posterior means and variance using the most current response and parameter estimates, a preferable approach that is data-driven yet still can be implemented on the fly. The dynamic nature of ability change poses an extra challenge in the estimation procedure when the item parameters are also unknown. In IRT, item parameters are usually computed by the MMLE/EM algorithm which assumes a population distribution that



(a) 20-item sessions

(b) 30-item sessions

Fig. 9 Trace plot of the estimates and the standard deviation of a randomly chosen student's ability under the MPWI with 50% pre-calibrated items condition. **a** 20-item sessions, **b** 30-item sessions

Table 2 Item exposure statistics comparison

Session length	Item selection	Pre-calibrated item proportions	New items		Pre-calibrated items	
			Mean	SD	Mean	SD
20	Random	20%	0.160	0.007	0.160	0.007
		50%	0.160	0.007	0.160	0.007
	Matched-difficulty	20%	0.159	0.037	0.164	0.031
		50%	0.158	0.035	0.162	0.029
	MPWI	20%	0.156	0.095	0.176	0.099
		50%	0.152	0.095	0.168	0.097
30	Random	20%	0.240	0.009	0.240	0.009
		50%	0.240	0.009	0.240	0.009
	Matched-difficulty	20%	0.239	0.039	0.243	0.033
		50%	0.238	0.036	0.242	0.026
	MPWI	20%	0.239	0.084	0.245	0.089
		50%	0.237	0.084	0.243	0.086

is integrated out in the marginalized likelihood function. Oftentimes there is no issue when the abilities remain constant, but it can become problematic when the same set of items is calibrated on different samples or the same sample with changing abilities (Wang et al., 2020). Similarly, for the Bayesian update rules, simultaneous estimation of item and ability parameters while the latter are constantly changing is also problematic because the starting prior of ability parameters would become mis-specified after ability changes, which in turn would lead to biased item calibration. There are two ways to rectify the issue: either to update the prior based on true ability changes or to fix the parameters of some of the items. Pilot simulations had shown that both methods would significantly improve parameter recovery, but in practice there is no way of knowing the true ability changes at the population level. Therefore, in this simulation the fixed parameter approach was adopted with varying proportions of pre-calibrated items, and the scaling issue was solved to make dynamic ability tracking possible again.

The simulation studies provided some preliminary support for the Bayesian update algorithm applied in an online learning system, but they still represented theoretical and oversimplified situations and had the following limitations. First, the change pattern simulated was stepwise and only occurred between sessions. In reality, there could be other possible change trajectories, such as linear growth within session as a function of item position or time (Eggen, 2012; Park et al., 2019b). Second, the learner ability in this simulation was assumed to be a unidimensional trait. In practice, learning materials are usually divided into multiple domains or areas. For instance, in the original paper on Math Garden (Klinkenberg et al., 2011), each item belonged to one of the four domains, which corresponded to the four algebraic operations (addition, subtraction, multiplication, and division). The ERS used in their study, however, did not differentiate between the four domains and only modeled the general math ability. Third, the change of abilities across sessions was drawn from $N\left(0.2, .0.1^2\right)$ in the current study, which was a relatively small change. It would also be interesting to explore whether the current results could be generalized to larger ability change conditions (i.e., $N(1,.0.5^2)$).

There are some other factors that could compromise the performance of the algorithm in the adaptive learning system. First, it was clear that there is an upward trend in the bias and RMSE of ability estimates over sessions, especially when most of the items in the pool are uncalibrated. It is unknown based on the results whether the bias and RMSE would continue to increase or level off after a few more sessions, but it is obvious that the accumulation of error would have a negative effect on the measurement performance in later sessions, which matters more than earlier sessions since they represented the learners' most current abilities. It has been shown that having more pre-calibrated items ameliorated the upward trend to some extent. Another potential fix to the issue is to apply more sophisticated estimation methods offline after a few sessions and reset the prior distributions accordingly. These limitations could be explored in future studies. Second, we assume the two-parameter logistic (2PL) model fits the data well. However, this assumption is likely violated in practice, which will compromise the precision of the algorithms. Hence, the current study can be extended in two directions. First, one may consider adding a real-time item-fit evaluation to continuously monitor model data fit. This requires the fit index to handle partial data and be computationally efficient. Second, one may resort to emergent deep learning algorithms (Piech et al., 2015) such as recurrent neural networks which can be considered a model-free approach. Third, in our simulations, we set τ to be 0.5 and noted that any values between 0.25 and 0.75 worked similarly well. However, the appropriate choice of τ certainly demands more research, as it is likely that a reasonable τ depends on the actual ability changes, in which a larger τ may be needed for a larger change. In addition, although we focus on item and ability parameter recovery in this study, how the choice of τ will affect the posterior standard deviation of estimated ability is worth exploring in the future as well, especially when the standard deviation will be used for stopping criterion in a variable-length CAT.

Assessment for learning is an emerging field of research that draws much attention from academic research and the testing industry. The combination of Bayesian update rule and MPWI as a psychometric tool holds great promise for the design of online learning systems that make instruction and learning in the future more intelligent, engaging, and productive. In addition, the methods introduced herein can also be used for designing adaptive psychological and neuroscientific experiments to maximize efficiency (Doebler, 2012).

Appendix for "On-The-Fly Parameter Estimation Based on Item Response Theory in Item-based Adaptive Learning Systems"

In this appendix, we present the derivations for the update formula for updating the posterior means and variance of α_j, β_j, and θ_i. Let the prior distributions be $\alpha_j \sim N\left(\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)$, $\beta_j \sim N\left(\mu_{\beta_j}, \sigma^2_{\beta_j}\right)$, and $\theta_i \sim N\left(\mu_{\theta_i}, \sigma^2_{\theta_i}\right)$ for person i and item j. Also let $z = \left(\alpha^*_j, \beta^*_j, \theta^*_i\right)^T$ be a standardized vector where $\alpha^*_j = \frac{\alpha_j - \mu_{\alpha_j}}{\sigma_{\alpha_j}}$, $\beta^*_j = \frac{\beta_j - \mu_{\beta_j}}{\sigma_{\beta_j}}$, and $\theta^*_i = \frac{\theta_i - \mu_{\theta_i}}{\sigma_{\theta_i}}$. The likelihood function for a single response based on a 2PL IRT model is

$$f\left(u_{ij}|z\right) = \Phi\left(\beta_j\theta_i + \alpha_j\right)^{u_{ij}} \left(1 - \Phi\left(\beta_j\theta_i + \alpha_j\right)\right)^{1-u_{ij}}, \tag{A1}$$

where $u_{ij} = 0$ or 1, Φ is a cumulative normal function. Take α_j as an example. When $u_{ij} = 1$, the derivatives in the expectations of the Equations 9 and 10 are, applying the chain rule,

$$\frac{\partial \log f\left(u_{ij}|z\right)}{\partial \alpha^*_j} = \sigma_{\alpha_j} \frac{\phi(x)}{\Phi(x)} \tag{A2}$$

$$\frac{\nabla^2 f\left(u_{ij}|\alpha^*_j\right)}{f\left(u_{ij}|\alpha^*_j\right)} = -\sigma^2_{\alpha_j} \frac{x\phi(x)}{\Phi(x)} \tag{A3}$$

where $x = \beta_j\theta_i + \alpha_j$, and $\phi(\cdot)$ is the normal density with given mean and variance. Then

$$E\left(\frac{\partial \log f\left(u_{ij}|z\right)}{\partial \alpha^*_j}\middle| u_{ij}\right) = \sigma_{\alpha_j} E\left(\frac{\phi(x)}{\Phi(x)}\middle| u_{ij}\right) \tag{A4}$$

Expand the expectation on the right as a triple integral (note that we update the parameter after observing u_{ij}, so below we write the equation as conditioning on u_{ij} rather than u.

$$E\left(\frac{\phi(x)}{\Phi(x)}\middle| u_{ij}\right) = \int\int\int \frac{\phi(x)}{\Phi(x)} p\left(\alpha_j, \beta_j, \theta_i | u_{ij}\right) d\alpha_j d\beta_j d\theta_i \tag{A5}$$

Because

$$p\left(\alpha_j, \beta_j, \theta_i | u_{ij}\right) = \frac{\Phi(x)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)\phi\left(\beta_j|\mu_{\beta_j}, \sigma^2_{\beta_j}\right)\phi\left(\theta_i|\mu_{\theta_i}, \sigma^2_{\theta_i}\right)}{\int\int\int \Phi(x)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)\phi\left(\beta_j|\mu_{\beta_j}, \sigma^2_{\beta_j}\right)\phi\left(\theta_i|\mu_{\theta_i}, \sigma^2_{\theta_i}\right) d\alpha_j d\beta_j d\theta_i}$$

Plugging this equation into (A5), we have,

$$E\left(\frac{\phi(x)}{\Phi(x)}\middle| u_{ij}\right) = \frac{\int\int\int \phi(x)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)\phi\left(\beta_j|\mu_{\beta_j}, \sigma^2_{\beta_j}\right)\phi\left(\theta_i|\mu_{\theta_i}, \sigma^2_{\theta_i}\right) d\alpha_j d\beta_j d\theta_i}{\int\int\int \Phi(x)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)\phi\left(\beta_j|\mu_{\beta_j}, \sigma^2_{\beta_j}\right)\phi\left(\theta_i|\mu_{\theta_i}, \sigma^2_{\theta_i}\right) d\alpha_j d\beta_j d\theta_i}$$

Now the problem can then be broken down into simplifying individual integral such as

$$\int\int\int \phi(x)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right)\phi\left(\beta_j|\mu_{\beta_j}, \sigma^2_{\beta_j}\right)\phi\left(\theta_i|\mu_{\theta_i}, \sigma^2_{\theta_i}\right) d\alpha_j d\beta_j d\theta_i$$

which is actually the integral of the product of 4 normal densities. It can be easily shown that the identity

$$\int \phi\left(x|\mu_x, \sigma^2_x\right)\phi\left(\mu_x|\mu_0, \sigma^2_0\right) d\mu_x = \phi\left(x|\mu_0, \sigma^2_x + \sigma^2_0\right)$$

holds. Write $\phi(x) = \phi(\beta_j\theta_i + \alpha_j) = \phi(-\beta_j\theta_i | \alpha_j, 1)$ and apply the above identity to simplify the integral. For example,

$$\int \phi\left(-\beta_j\theta_i|\alpha_j, 1\right)\phi\left(\alpha_j|\mu_{\alpha_j}, \sigma^2_{\alpha_j}\right) d\alpha_j = \phi\left(-\beta_j\theta_i|\mu_{\alpha_j}, 1 + \sigma^2_{\alpha_j}\right)$$

The identity can be applied three times, and for the last time it is necessary to replace θ_i with μ_{θ_i} in the variance component of the normal density. The result of the simplification is a new normal PDF

$$\phi\left(-\mu_{\alpha_j}|\mu_{\beta_j}\mu_{\theta_j}, 1 + \sigma^2_{\alpha_j} + \sigma^2_{\beta_j}\mu^2_{\theta_i} + \sigma^2_{\theta_i}\mu^2_{\beta_j}\right)$$

$$= \frac{1}{\sqrt{1 + \sigma^2_{\alpha_j} + \sigma^2_{\beta_j}\mu^2_{\theta_i} + \sigma^2_{\theta_i}\mu^2_{\beta_j}}} \phi\left(\frac{\mu_{\alpha_j} + \mu_{\beta_j}\mu_{\theta_i}}{\sqrt{1 + \sigma^2_{\alpha_j} + \sigma^2_{\beta_j}\mu^2_{\theta_i} + \sigma^2_{\theta_i}\mu^2_{\beta_j}}}\right)$$

Apply similar procedures to the denominator (there is a similar identity for the product of a cumulative normal density as well) and replace θ_i with μ_{θ_i} when needed, the expectation can be approximated as

$$E\left(\frac{\phi(x)}{\Phi(x)}\right) \approx \frac{1}{v}\frac{\phi(x^*)}{\Phi(x^*)} \tag{A6}$$

where

$$v = \sqrt{1 + \sigma^2_{\alpha_j} + \sigma^2_{\beta_j}\mu^2_{\theta_i} + \sigma^2_{\theta_i}\mu^2_{\beta_j}} \tag{A7}$$

$$x^* = \frac{\mu_{\alpha_j} + \mu_{\beta_j}\mu_{\theta_i}}{\nu} \tag{A8}$$

The expectations for β_j and θ_i can be found analogously, and the update rules for the posterior means and variances of item and ability parameters are given below

$$\mu'_{\alpha_j} = \mu_{\alpha_j} + \frac{\sigma^2_{\alpha_j}}{\nu}\frac{\phi(x^*)}{\Phi(x^*)} \tag{A9}$$

$$\mu'_{\beta_j} = \mu_{\beta_j} + \frac{\mu_{\theta_i}\sigma^2_{\beta_j}}{\nu}\frac{\phi(x^*)}{\Phi(x^*)} \tag{A10}$$

$$\mu'_{\theta_i} = \mu_{\theta_i} + \frac{\mu_{\beta_j}\sigma^2_{\theta_i}}{\nu}\frac{\phi(x^*)}{\Phi(x^*)} \tag{A11}$$

$$\sigma^{2\prime}_{\alpha_j} = \sigma^2_{\alpha_j}\max\left(1 - \left(\frac{\sigma_{\alpha_j}}{\nu}\right)^2\left[\frac{x^{*(m)}\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\Phi\left(x^{*(m)}\right)^2}\right], \kappa\right) \tag{A12}$$

$$\sigma^{2\prime}_{\beta_j} = \sigma^2_{\beta_j}\max\left(1 - \left(\frac{\mu_{\theta_i}\sigma_{\beta_j}}{\nu}\right)^2\left[\frac{x^{*(m)}\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\Phi\left(x^{*(m)}\right)^2}\right], \kappa\right) \tag{A13}$$

$$\sigma^{2\prime}_{\theta_i} = \sigma^2_{\theta_i}\max\left(1 - \left(\frac{\mu_{\beta_j}\sigma_{\theta_i}}{\nu}\right)^2\left[\frac{x^{*(m)}\phi\left(x^{*(m)}\right)}{\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\Phi\left(x^{*(m)}\right)^2}\right], \kappa\right) \tag{A14}$$

where κ is a constant slightly larger than 0 to make sure that posterior variance is always positive. When $u_{ij}=0$, Equation A2 and A3 change to:

$$\frac{\partial \log f\left(u_{ij}|z\right)}{\partial \alpha^*_j} = -\sigma_{\alpha_j}\frac{\phi(x)}{1-\Phi(x)} \tag{A15}$$

$$\frac{\nabla^2 f\left(u_{ij}|\alpha^*_j\right)}{f\left(u_{ij}|\alpha^*_j\right)} = \sigma^2_{\alpha_j}\frac{x\phi(x)}{1-\Phi(x)} \tag{A16}$$

Using the similar procedure, the update rules for the posterior means and variances of item and ability parameters are given below

$$\mu'_{\alpha_j} = \mu_{\alpha_j} - \frac{\sigma^2_{\alpha_j}}{\nu}\frac{\phi(x^*)}{1-\Phi(x^*)} \tag{A17}$$

$$\mu'_{\beta_j} = \mu_{\beta_j} - \frac{\mu_{\theta_i}\sigma^2_{\beta_j}}{\nu}\frac{\phi(x^*)}{1-\Phi(x^*)} \tag{A18}$$

$$\mu'_{\theta_i} = \mu_{\theta_i} - \frac{\mu_{\beta_j}\sigma^2_{\theta_i}}{\nu}\frac{\phi(x^*)}{1-\Phi(x^*)} \tag{A19}$$

$$\sigma^{2\prime}_{\alpha_j} = \sigma^2_{\alpha_j}\max\left(1 - \left(\frac{\sigma_{\alpha_j}}{\nu}\right)^2\left[\frac{-x^{*(m)}\phi\left(x^{*(m)}\right)}{1-\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\left(1-\Phi\left(x^{*(m)}\right)\right)^2}\right], \kappa\right) \tag{A20}$$

$$\sigma^{2\prime}_{\beta_j} = \sigma^2_{\beta_j}\max\left(1 - \left(\frac{\mu_{\theta_i}\sigma_{\beta_j}}{\nu}\right)^2\left[\frac{-x^{*(m)}\phi\left(x^{*(m)}\right)}{1-\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\left(1-\Phi\left(x^{*(m)}\right)\right)^2}\right], \kappa\right) \tag{A21}$$

$$\sigma^{2\prime}_{\theta_i} = \sigma^2_{\theta_i}\max\left(1 - \left(\frac{\mu_{\beta_j}\sigma_{\theta_i}}{\nu}\right)^2\left[\frac{-x^{*(m)}\phi\left(x^{*(m)}\right)}{1-\Phi\left(x^{*(m)}\right)} + \frac{\phi\left(x^{*(m)}\right)^2}{\left(1-\Phi\left(x^{*(m)}\right)\right)^2}\right], \kappa\right) \tag{A22}$$

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Data availability Source code available at https://github.com/wang4066/On-the-Fly-Calibration.

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Open practices statement

None of the data or materials for the experiments reported here are available, and none of the experiments was preregistered, because the entire study is based on simulated data. The source code for the entire project is available on GitHub.

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